

Beyond Single-Exponential Self-Improvement: Takeoff Kernels, Response Modes, and Source Identification

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Working draft, July 2026

Abstract

A takeoff kernel records when eventual improvement is delivered; it need not be exponential. We use this general response object to recast a linearized solver–verifier theory as a joint rank-one null: normalized solver, verifier, and gap trajectories must share one discrete mode. A finite-state extension produces exponential response modes, but decomposing an observed kernel into such modes is a source-separation problem, and response modes do not by themselves identify mechanisms. Synthetic studies confirm that rejecting rank one is easier than recovering endpoints or modes. In figure-derived Phi-4 curves, one exact shared rate fails, while power-law relaxation often matches or exceeds finite-modal forecasts and a shared phase-change model does not win. A resource-scaled reproduction closes the per-token gap without a stable modal law. Dense checkpoints further resolve one-optimizer-update changes coupled to response length and accuracy, not a smooth capability takeoff. The evidence supports kernel-based response diagnosis and rejection of a global shared-rate law, but not a discrete mode count, a universal local exponential, or identified causal sources.

1 Question and Scope

A standard self-improvement loop generates candidate responses, evaluates or filters them with a verifier, and trains the solver on the resulting signal [1, 2]. Recent work by Sun et al. models the within-run training trajectory through coupled solver and verifier uncertainties [3]. Its linearized theory predicts exponential convergence and an intrinsic limit. The present question is narrower than a general theory of recursive intelligence: what finite response shape does this model predict, and how can that prediction be tested or relaxed?

The answer begins with a distinction between two clocks. The *inner clock* indexes optimization time or checkpoints within a training run. The *outer clock* indexes complete generate–verify–retrain rounds. The theory analyzed below concerns the inner clock. Applying the same kernel language to the outer clock is possible, but the corresponding modes have a different meaning and should be estimated separately.

The takeoff kernel itself makes no exponential assumption. It is the sequence of normalized checkpoint-to-checkpoint gains, and it can be concentrated, long-tailed, signed, piecewise, or irregular. Exponentials enter only when a particular dynamical model is imposed. An exponential curve approaching a

limit has a geometric residual, constant increment ratios, and rank-one Hankel matrices. For the coupled solver–verifier law, solver, verifier, and gap must also share the same geometric mode. This turns a visual curve-fitting claim into a joint, falsifiable null without confusing the null with the general kernel framework.

The theoretical development separates the response object from models of its sources. We first normalize increasing scores and decreasing uncertainties on a common scale, then derive the rank-one restrictions implied by the standard law. A finite-dimensional hidden state gives one tractable subclass in which active modes determine settling time and noiseless Hankel rank records modal count. Recovering those temporal components is already an inverse problem; assigning them to solver learning, verifier drift, data refresh, or optimizer state is a further source-identification problem. A rank-one versus rank-two construction isolates the basic point: identical endpoints need not imply similar finite takeoff.

The empirical half asks what can actually be learned from short, noisy trajectories. Synthetic studies first separate rejection of rank one from the harder tasks of estimating the asymptote and recovering individual modes. We then examine vector-extracted Phi-4 trajectories from Sun et al., run a resource-scaled fresh-seed reproduction, compare global modal, time-local, and nonmodal forecasts, and repeat one seed with quarter-epoch measurements through epoch four. The published curves are inconsistent with one exact shared rate, but neither they nor the reproduction identify a stable replacement mode count or local exponential. The intended outcome is therefore a sharper inverse problem and measurement protocol, not a claim that hidden mechanisms have been recovered.

2 Response Normalization

Let Y_n be a scalar observable measured at equally spaced checkpoints and suppose $Y_n \rightarrow Y_\infty$ with $Y_0 \neq Y_\infty$. The observable may be an increasing performance score or a decreasing uncertainty. The following normalization treats both cases identically.

Definition 1 (Residual, profile, and takeoff kernel). *Define*

$$e_n = \frac{Y_n - Y_\infty}{Y_0 - Y_\infty}, \quad F_n = 1 - e_n, \quad (1)$$

and, for $n \geq 1$,

$$\kappa_n = F_n - F_{n-1} = e_{n-1} - e_n. \quad (2)$$

The sequence e_n is the normalized residual distance to the limit, F_n is the cumulative realized response, and κ_n is the response delivered at checkpoint n .

By telescoping,

$$\sum_{j=1}^N \kappa_j = F_N = 1 - e_N. \quad (3)$$

Hence $\sum_{j \geq 1} \kappa_j = 1$ whenever $e_n \rightarrow 0$. If F_n is monotone, the kernel is nonnegative and can be read as a distribution of improvement over checkpoints. Negative entries instead mark regression, overshoot, or ringing. With this common scale in place, a single exponential becomes an exact algebraic hypothesis rather than a qualitative curve shape.

Remark 1 (The kernel precedes its parameterization). *Equations (1)–(2) define a kernel for any convergent observed trajectory once an endpoint and clock are fixed. They do not require linearity, smoothness, or an exponential mixture. Finite modal, power-law, phase-changing, and nonparametric laws are competing descriptions of the kernel, not competing definitions of it. Consequently, the kernel formalism alone is descriptive; empirical content comes from restrictions that forecast untouched checkpoints and replicate across seeds.*

3 The Standard Law Is Rank One

The first result identifies exactly what a single exponential asserts about the finite response.

Theorem 1 (Single-mode equivalence). *Let $e_0 = 1$, let $e_n \rightarrow 0$, and define $\kappa_n = e_{n-1} - e_n$ for $n \geq 1$. For any $\lambda \in \mathbb{C}$ with $|\lambda| < 1$, the following are equivalent:*

$$e_n = \lambda^n \quad (n \geq 0), \quad (4)$$

$$\kappa_n = (1 - \lambda)\lambda^{n-1} \quad (n \geq 1). \quad (5)$$

When the response is real and monotone, $\lambda \in [0, 1]$. Moreover, wherever the ratios are defined,

$$\frac{\kappa_{n+1}}{\kappa_n} = \lambda, \quad (6)$$

and every adjacent second-order Hankel minor vanishes:

$$\kappa_{n+1}\kappa_{n-1} - \kappa_n^2 = 0. \quad (7)$$

Proof. If $e_n = \lambda^n$, then

$$\kappa_n = e_{n-1} - e_n = \lambda^{n-1} - \lambda^n = (1 - \lambda)\lambda^{n-1}.$$

Conversely, if (5) holds, telescoping gives

$$F_n = \sum_{j=1}^n (1 - \lambda)\lambda^{j-1} = 1 - \lambda^n.$$

Since $e_n = 1 - F_n$, equation (4) follows. Equations (6) and (7) follow by direct substitution. If the real response is monotone, then $0 \leq e_{n+1} \leq e_n$ and therefore $\lambda \in [0, 1]$. $\square$

The theorem replaces a visual statement about curve shape by local algebraic restrictions. A high in-sample coefficient of determination for an exponential fit does not by itself test those restrictions, especially when the asymptote is also estimated.

3.1 Application to solver–verifier dynamics

Sun et al. define solver uncertainty $U_s(t)$, verifier uncertainty $U_v(t)$, and gap

$$G(t) = U_s(t) - U_v(t). \quad (8)$$

Their phenomenological dynamics are

$$\frac{dU_s}{dt} = -\alpha E(t), \quad \frac{dU_v}{dt} = -\beta E(t), \quad \alpha > \beta \geq 0, \quad (9)$$

where $E(t) = f(G(t))$. Around an initial state they use the affine approximation

$$E(t) \approx kG(t) - b. \quad (10)$$

Writing $G_\infty = b/k$ and $\gamma = k(\alpha - \beta) > 0$, their solution has the form

$$G(t) - G_\infty = C_G e^{-\gamma t}, \quad (11)$$

$$U_s(t) - U_{s,\infty} = C_s e^{-\gamma t}, \quad (12)$$

$$U_v(t) - U_{v,\infty} = C_v e^{-\gamma t}, \quad (13)$$

for constants determined by the initial condition.

Corollary 1 (Kernel form of the solver–verifier law). *Sample (11)–(13) at times $t_n = n\Delta$ with $\Delta > 0$. For every nonconstant observable $X \in \{G, U_s, U_v\}$, its normalized residual and takeoff kernel are*

$$e_n^{(X)} = \lambda^n, \quad \kappa_n^{(X)} = (1 - \lambda)\lambda^{n-1}, \quad \lambda = e^{-k(\alpha - \beta)\Delta}. \quad (14)$$

Thus the three normalized trajectories share one mode, and each exact kernel Hankel matrix has rank one.

Proof. For a nonconstant X , equations (11)–(13) give

$$\frac{X(t_n) - X_\infty}{X(0) - X_\infty} = e^{-\gamma n\Delta} = \lambda^n.$$

Apply Theorem 1. The sequence $\kappa_n = (1 - \lambda)\lambda^{n-1}$ is an outer product on any Hankel grid, so its nonzero Hankel matrix has rank one. $\square$

Corollary 2 (Joint empirical null). *Under (9)–(10), equally spaced observations obey all of the following restrictions:*

1. $e_n^{(s)} = e_n^{(v)} = e_n^{(G)}$ whenever the three normalizations exist;
2. each kernel has the same constant ratio λ ;
3. every second-order kernel Hankel minor vanishes;
4. if $\beta > 0$, then

$$\frac{U_s(t) - U_{s,\infty}}{U_v(t) - U_{v,\infty}} = \frac{\alpha}{\beta}$$

throughout the nonstationary trajectory.

Proof. The first three statements follow from Corollary 1. For the fourth, integrating (9) from t to infinity gives

$$U_s(t) - U_{s,\infty} = \alpha \int_t^\infty E(\tau) d\tau, \\ U_v(t) - U_{v,\infty} = \beta \int_t^\infty E(\tau) d\tau.$$

Taking the ratio proves the claim. $\square$

Corollary 2 is the first result to test empirically. It is strictly sharper than fitting one exponential to each series independently: it tests whether all observables arise from the same one-dimensional active response.

4 Finite-Dimensional Modal Generalization

The rank-one null can be relaxed without abandoning a tractable response law, but this is a model class for the observed response, not a definition of the kernel and not yet a claim about its physical sources. Let $x_n \in \mathbb{C}^d$ be a hidden state centered at its limiting value and suppose that, over the observation window, it evolves linearly:

$$x_{n+1} = Ax_n. \quad (15)$$

Let the centered scalar observable be

$$y_n = Y_n - Y_\infty = c^* x_n. \quad (16)$$

Theorem 2 (Modal takeoff representation). *Assume A is diagonalizable over $\mathbb{C}$. Let $\theta_1, \dots, \theta_R$ be the distinct eigenvalues that are active for the initial state x_0 and observation vector c , and assume $|\theta_j| < 1$. Then there are nonzero modal amplitudes $a_1, \dots, a_R$ such that*

$$y_n = \sum_{j=1}^R a_j \theta_j^n. \quad (17)$$

If $y_0 = \sum_j a_j \neq 0$, the normalized residual and takeoff kernel are

$$e_n = \frac{\sum_{j=1}^R a_j \theta_j^n}{\sum_{j=1}^R a_j}, \quad (18)$$

$$\kappa_n = \frac{\sum_{j=1}^R a_j (1 - \theta_j) \theta_j^{n-1}}{\sum_{j=1}^R a_j}. \quad (19)$$

For real A, c, x_0 , nonreal modes and amplitudes occur in conjugate pairs, so e_n and κ_n remain real.

Proof. Diagonalizability gives right eigenvectors v_j and spectral projectors P_j such that

$$A^n = \sum_j \theta_j^n P_j.$$

Substituting this expansion into (16) gives

$$y_n = c^* A^n x_0 = \sum_j (c^* P_j x_0) \theta_j^n.$$

Discard terms with zero coefficient and combine repeated eigenvalues. This produces (17) with the distinct active modes. Since $y_0 = \sum_j a_j$, division by y_0 gives (18). Taking the difference $e_{n-1} - e_n$ gives (19). For real data, the real canonical form pairs every nonreal eigenvalue with its complex conjugate, and the corresponding terms sum to a real sequence. $\square$

Remark 2 (What the standard model suppresses). *After centering on its trajectory-specific limit, the linearized solver-verifier system has matrix*

$$B = -k \begin{pmatrix} \alpha & -\alpha \\ \beta & -\beta \end{pmatrix}. \quad (20)$$

Its eigenvalues are 0 and $-k(\alpha - \beta)$. The zero direction parametrizes the continuum of stationary endpoints and vanishes after centering on the endpoint selected by the initial condition. Only the decaying direction is active, which explains rank one. A general stable two-state Jacobian has two decaying eigenvalues; sampling it at interval Δ yields modes $\theta_j = e^{\mu_j \Delta}$ and hence a rank-at-most-two modal kernel.

Theorem 2 does not identify mechanisms from a curve. It identifies response modes. Assigning a recovered fast or slow mode to solver learning, verifier drift, data refresh, or optimizer state requires additional measurements or interventions.

4.1 Response modes versus latent sources

The distinction can be written as a small source-separation problem. Suppose the centered measured observables are

$$y_n = Cz_n, \quad z_{n+1} = Az_n, \quad (21)$$

where the coordinates of z_n are latent response channels and C is an unknown observation map. Diagonalizing A gives temporal response modes in each measured curve, but it does not determine the columns of C or attach them to named training mechanisms. Different latent systems can therefore produce the same observed kernel.

In the solver-verifier data, this ambiguity is particularly sharp. The gap is derived, $G = U_s - U_v$, so the three plotted curves are not three independent sensors. A fitted second mode is evidence for a second effective timescale in the response model; it is not, by itself, evidence for a second causal source. Source separation requires additional independent observables, repeated conditions with different mixing weights, or interventions that selectively perturb a channel. The empirical analyses below therefore separate response diagnosis from mechanistic attribution.

5 Modal Radius and Settling

The modal representation tells us what components are present. We next ask what those components imply for the time required to approach the endpoint.

Definition 2 (Settling time). *For $\epsilon > 0$, define*

$$T_\epsilon = \min\{N \geq 0 : |e_n| \leq \epsilon \text{ for every } n \geq N\}. \quad (22)$$

The modal radius must refer to modes active in the measured observable, not to every eigenvalue of a latent transition matrix.

Theorem 3 (Modal radius controls settling). *Under the hypotheses of Theorem 2, write $d_j = a_j / \sum_\ell a_\ell$ and let*

$$\lambda_\star = \max_{1 \leq j \leq R} |\theta_j|, \quad 0 < \lambda_\star < 1. \quad (23)$$

There are constants $0 < c \leq C < \infty$ such that, for every $n \geq 0$,

$$|e_n| \leq C\lambda_\star^n, \quad (24)$$

$$\max_{0 \leq \ell < R} |e_{n+\ell}| \geq c\lambda_\star^n. \quad (25)$$

Consequently, for all sufficiently small $\epsilon > 0$,

$$\frac{\log(c/\epsilon)}{-\log \lambda_\star} \leq T_\epsilon \leq \left\lceil \frac{\log(C/\epsilon)}{-\log \lambda_\star} \right\rceil, \quad (26)$$

and hence

$$\lim_{\epsilon \downarrow 0} \frac{T_\epsilon}{\log(1/\epsilon)} = \frac{1}{-\log \lambda_\star}. \quad (27)$$

Proof. Equation (18) gives

$$|e_n| \leq \sum_{j=1}^R |d_j| |\theta_j|^n \leq \left(\sum_{j=1}^R |d_j| \right) \lambda_\star^n,$$

so (24) holds with $C = \sum_j |d_j|$.

For the lower bound, let V be the square Vandermonde matrix

$$V_{\ell j} = \theta_j^\ell, \quad 0 \leq \ell < R, \quad 1 \leq j \leq R.$$

The modes are distinct, so V is invertible. Define $q_n = (d_1 \theta_1^n, \dots, d_R \theta_R^n)^\top$. The block of R successive residuals is

$$(e_n, \dots, e_{n+R-1})^\top = V q_n.$$

Let $\sigma_{\min}(V) > 0$ be the smallest singular value. Since at least one active coefficient d_j has $|\theta_j| = \lambda_\star$,

$$\max_{0 \leq \ell < R} |e_{n+\ell}| \geq \frac{\sigma_{\min}(V)}{\sqrt{R}} \|q_n\|_2 \geq c\lambda_\star^n$$

for a fixed $c > 0$. This proves (25).

The upper bound in (26) follows because every $m \geq N$ obeys $|e_m| \leq C\lambda_\star^N$. If the trajectory has settled by T_ϵ , every member of its next length- R block is at most ϵ , so (25) gives $c\lambda_\star^{T_\epsilon} \leq \epsilon$, which is the lower bound. Dividing by $\log(1/\epsilon)$ and sending $\epsilon \downarrow 0$ proves (27). $\square$

Remark 3 (Cancellation and visible modes). *An eigenvalue of A is absent when its coefficient $a_j = c^* P_j x_0$ vanishes. Such a mode is unexcited by the initial condition, unobserved by the measurement, or both; it cannot be inferred from that scalar trajectory. Equal-radius active modes can also cancel at isolated checkpoints. The block lower bound in Theorem 3 shows why those cancellations do not change the asymptotic settling exponent.*

Remark 4 (Shape classes). *If $\theta_j \in [0, 1)$ and $d_j \geq 0$ for every j , then*

$$\kappa_n = \sum_j d_j (1 - \theta_j) \theta_j^{n-1} \geq 0,$$

so the response is monotone. Negative real modes can alternate, while nonreal conjugate pairs produce damped ringing. These are implications of the fitted response modes, not yet causal labels for training mechanisms.

Remark 5 (Jordan blocks and repeated modes). *If A is not diagonalizable, a Jordan block replaces a pure exponential by $p_j(n)\theta_j^n$, where p_j is a polynomial whose degree is smaller than the block size. For largest active block size m ,*

$$|e_n| \leq Cn^{m-1}\lambda_\star^n.$$

The leading settling scale remains $\log(1/\epsilon)/(-\log \lambda_\star)$, with at most an additional order- $\log \log(1/\epsilon)$ correction. Thus a repeated mode is not correctly represented by simply listing the same eigenvalue several times in Theorem 2.

Remark 6 (Algebraic tails are outside the finite-modal model). *If $e_n \sim cn^{-p}$ with $c \neq 0$ and $p > 0$, then $|e_n|^{1/n} \rightarrow 1$ and T_ϵ grows on the order of $\epsilon^{-1/p}$. No finite sum of stable polynomial-exponential modes can equal such a tail eventually, because every such sum has root-decay rate strictly below one. An observed algebraic tail therefore points to time-varying, infinite-dimensional, or broad-spectrum dynamics rather than a larger but finite stable modal order.*

For completely monotone residuals, a broad spectrum can be represented as a positive mixture of exponentials,

$$e(t) = \int_0^\infty e^{-\lambda t} d\mu(\lambda), \quad (28)$$

with a finite-modal response corresponding to an atomic measure μ . This representation is a useful comparison class, not a universal representation of arbitrary signed or phase-changing kernels. A power-law forecast can thus be a compact description of many timescales without identifying a finite list of modes.

6 Hankel Rank and Exact Modal Recovery

Settling is the forward consequence of the active modes. The corresponding inverse question is whether those modes can be counted and recovered from the observed response.

Write the modal kernel from (19) as

$$\kappa_n = \sum_{j=1}^R b_j \theta_j^{n-1}, \quad b_j = d_j(1 - \theta_j). \quad (29)$$

Because $|\theta_j| < 1$, every active d_j gives a nonzero b_j .

Theorem 4 (Hankel rank equals active modal count). *Let $\theta_1, \dots, \theta_R$ be distinct and let $b_j \neq 0$. For $m \geq R$, form the square kernel Hankel matrix*

$$H_m = (\kappa_{i+j-1})_{i,j=0}^{m-1}. \quad (30)$$

Then

$$\text{rank } H_m = R. \quad (31)$$

In particular, the leading $R \times R$ determinant is

$$\det H_R = \left(\prod_{j=1}^R b_j \right) \left(\prod_{1 \leq p < q \leq R} (\theta_q - \theta_p)^2 \right) \neq 0. \quad (32)$$

Proof. Let V_m be the $m \times R$ Vandermonde matrix $(V_m)_{ij} = \theta_j^i$ for $0 \leq i < m$. Equation (29) gives the factorization

$$H_m = V_m \text{diag}(b_1, \dots, b_R) V_m^\top. \quad (33)$$

Thus $\text{rank } H_m \leq R$. When $m = R$, the Vandermonde determinant is $\det V_R = \prod_{p < q} (\theta_q - \theta_p)$, so (32) follows and H_R is nonsingular. For $m > R$, H_R is a nonsingular principal submatrix of H_m , giving $\text{rank } H_m \geq R$. Hence equality holds. $\square$

Corollary 3 (Noiseless matrix-pencil recovery). *Define*

$$H_0 = (\kappa_{i+j+1})_{i,j=0}^{R-1}, \quad H_1 = (\kappa_{i+j+2})_{i,j=0}^{R-1}. \quad (34)$$

Then the generalized eigenvalues of the pencil (H_1, H_0) are precisely $\theta_1, \dots, \theta_R$:

$$\det(H_1 - zH_0) = \det(V_R)^2 \left(\prod_{j=1}^R b_j \right) \prod_{j=1}^R (\theta_j - z). \quad (35)$$

Thus $2R$ consecutive kernel values suffice for exact modal recovery in the nondegenerate noiseless model.

Proof. With $\Theta = \text{diag}(\theta_1, \dots, \theta_R)$ and $D = \text{diag}(b_1, \dots, b_R)$, the two matrices factor as

$$H_0 = V_R D V_R^\top, \quad H_1 = V_R D \Theta V_R^\top.$$

Taking the determinant of $V_R D (\Theta - zI) V_R^\top$ proves (35). $\square$

Exact rank is not a literal noisy-data estimator: an arbitrarily small generic perturbation makes a finite Hankel matrix full rank. The empirical analogue must therefore use a singular-value spectrum, regularization, bootstrap stability, and held-out prediction. Theorem 4 supplies the population target for that procedure.

7 Same Endpoints, Different Takeoff

The recovery results make modal order observable in principle. The following construction shows why that order matters even when the usual endpoint comparison cannot distinguish two trajectories.

Corollary 4 (Endpoint separation). *Fix $Y_0 \neq Y_\infty$, $0 < a < b < 1$, and $w \in (0, 1)$. Define*

$$Y_n^{(1)} = Y_\infty + (Y_0 - Y_\infty) a^n, \quad (36)$$

$$Y_n^{(2)} = Y_\infty + (Y_0 - Y_\infty) (w a^n + (1 - w) b^n). \quad (37)$$

The trajectories have the same initial value, limit, and total normalized kernel mass. The first has Hankel rank one and modal radius a ; the second has Hankel rank two and modal radius b . Their kernels are

$$\kappa_n^{(1)} = (1 - a) a^{n-1}, \quad (38)$$

$$\kappa_n^{(2)} = w(1 - a) a^{n-1} + (1 - w)(1 - b) b^{n-1}. \quad (39)$$

Moreover, for $0 < \epsilon < 1 - w$,

$$T_\epsilon^{(1)} = \left\lceil \frac{\log \epsilon}{\log a} \right\rceil, \quad T_\epsilon^{(2)} \geq \frac{\log((1 - w)/\epsilon)}{-\log b}. \quad (40)$$

For fixed a, w, ϵ , the second settling time becomes arbitrarily large as $b \uparrow 1$, although the endpoints remain unchanged.

Proof. Both normalized residuals equal one at $n = 0$ and converge to zero. Taking successive differences gives (38)–(39); both kernels are nonnegative and telescope to total mass one. The rank statements follow from Theorem 4. The first settling formula follows from $a^n \leq \epsilon$. Since the second residual satisfies $e_n^{(2)} \geq (1 - w) b^n$, settling requires $(1 - w) b^n \leq \epsilon$, which gives the stated lower bound. $\square$

Corollary 4 is the first synthetic experiment. It shows exactly what endpoint or final-gain comparisons omit: the slow mode can carry modest weight while controlling the late response.

8 Synthetic Detectability

Exact Hankel rank is discontinuous under noise, so Theorem 4 does not by itself determine how many checkpoints an experiment needs. We estimate that requirement for the construction in Figure 1. The null residual is

$$e_n^{(0)} = 0.45^n, \quad (41)$$

and the rank-two alternative is

$$e_n^{(1)} = 0.75(0.45)^n + 0.25(0.94)^n. \quad (42)$$

At each checkpoint after $n = 0$, we add independent Gaussian noise with standard deviation σ on the normalized residual scale. The baseline $e_0 = 1$ and the limiting endpoint are treated as known in this first power study.

For N observed checkpoints, use the common square size $m = \lfloor N/2 \rfloor$, the largest size supported by the corresponding length- $(N - 1)$ kernel, and form an $m \times m$ Hankel matrix from either e_n or $\kappa_n = e_{n-1} - e_n$. The test statistic is the singular-value ratio

$$S = \frac{s_2(H)}{s_1(H)}. \quad (43)$$

An independent set of 2,000 rank-one simulations calibrates the upper five-percent threshold. A second null set checks the false-positive rate, and 2,000 rank-two simulations estimate power. The largest absolute false-positive calibration error across nonzero-noise conditions is 0.014.

The gap between the panels has a direct modal explanation. In the residual, the slow-mode amplitude is $d_{\text{slow}} = 0.25$. In the kernel it becomes

$$b_{\text{slow}} = d_{\text{slow}}(1 - \theta_{\text{slow}}) = 0.25(1 - 0.94) = 0.015.$$

Relative to the fast mode, differencing attenuates the slow component by the additional factor

$$\frac{1 - \theta_{\text{slow}}}{1 - \theta_{\text{fast}}} = \frac{0.06}{0.55} \approx 0.109, \quad (44)$$

A small slow mode controls late self-improvement

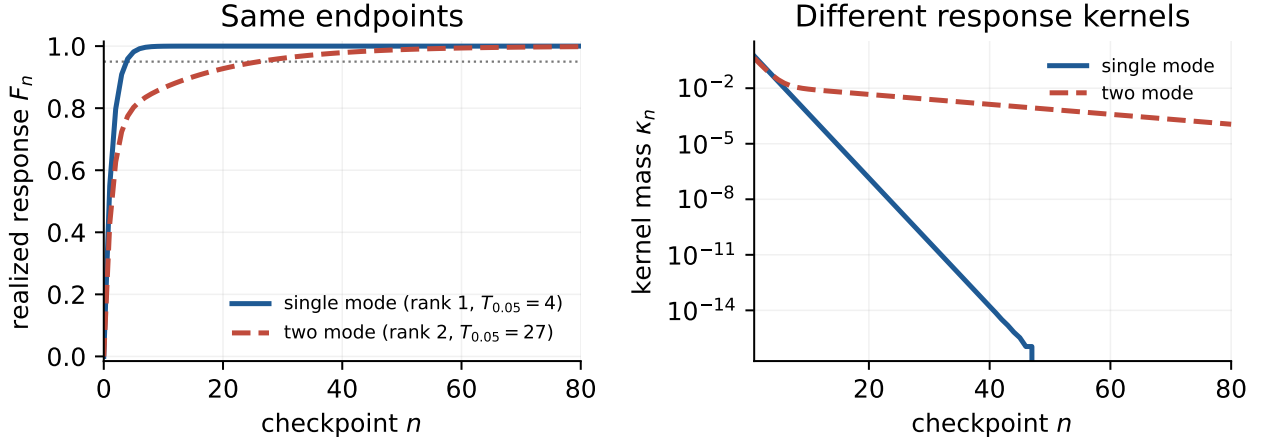
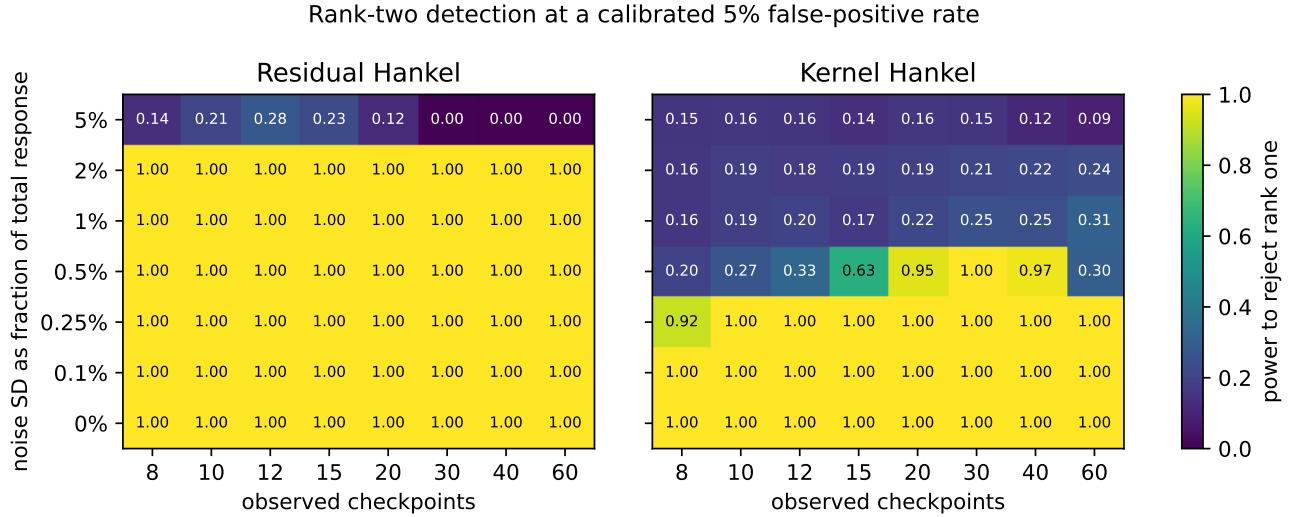


Figure 1: **Same endpoints, different response kernels.** The rank-one trajectory uses $a = 0.45$. The rank-two trajectory assigns weight $w = 0.75$ to that fast mode and weight 0.25 to a slow mode $b = 0.94$. Both profiles begin at zero, converge to one, and have total kernel mass one. Nevertheless, their exact Hankel ranks are one and two, and their $\epsilon = 0.05$ settling times are 4 and 27 checkpoints. The slow mode is lightly weighted in the early response but controls the tail.



Alternative: modes 0.45 and 0.94; slow-mode weight 0.25; 2000 Monte Carlo replicates per calibration/test sample.

Figure 2: **Power to detect the slow second mode.** Each cell reports the probability of rejecting rank one using the calibrated singular-value-ratio statistic (43). With the endpoint known, the residual Hankel detects the rank-two alternative with at least 0.8 power from eight checkpoints through $\sigma = 0.02$. Kernel Hankels are more demanding: eight checkpoints suffice through $\sigma = 0.0025$, while twenty are required at $\sigma = 0.005$; no tested count reaches 0.8 power at $\sigma \geq 0.01$. Enlarging an unregularized square Hankel can eventually add more noise-only tail than signal, so power need not be monotone in the plotted checkpoint count.

while independent observation noise acquires a differenced standard deviation of approximately $\sqrt{2}\sigma$. Residual Hankels are therefore preferable for detecting modal order when the asymptote is trustworthy; kernels remain the natural object for interpreting when improvement arrives.

These numbers are design guidance for this alternative, not a universal sample-complexity theorem. They are optimistic

about the residual because Y_∞ is known. An asymptote error introduces a nearly constant residual component and can mimic an additional mode. The next sensitivity analysis therefore profiles over Y_∞ and compares held-out modal fits. Varying slow-mode weight and separation, and comparing regularized or windowed Hankels, remain subsequent robustness checks.

9 Unknown Asymptote and Held-Out Forecasting

We next remove the known-endpoint advantage. On a training prefix, we fit the nested models

$$Y_n^{[1]} = Y_\infty + c\lambda^n, \quad (45)$$

$$Y_n^{[2]} = Y_\infty + c_1\theta_1^n + c_2\theta_2^n, \quad (46)$$

with $c, c_1, c_2 \geq 0$ and $Y_\infty \in [-0.25, 0.25]$. For each candidate mode tuple on a fixed grid, the endpoint and amplitudes are obtained by linear least squares; the grid search then selects the smallest training error. The last quarter of the checkpoints, with at least three held out, is used only for forecasting. Model preference is measured by

$$D = \log \frac{\text{MSE}_{\text{holdout}}^{[1]}}{\text{MSE}_{\text{holdout}}^{[2]}}. \quad (47)$$

For every noise level and checkpoint count, an independent set of 2,000 rank-one simulations calibrates the upper five-percent threshold for D ; separate null and rank-two sets estimate size and power. The true modes 0.45 and 0.94 are included in the grid, so this remains an intentionally favorable test of the proposed alternative. The largest absolute false-positive calibration error is 0.013.

The minimum tested counts attaining 0.8 power are 10, 10, 12, 15, 20, and 40 checkpoints at noise levels 0.1%, 0.25%, 0.5%, 1%, 2%, and 5% of the total response, respectively. These are detection thresholds, not parameter-recovery thresholds. At 1% noise, for example, power is already 0.996 at twenty checkpoints, yet the median endpoint error is 8.00% and both modes are recovered within 0.02 in only 0.262 of simulations. At sixty checkpoints those figures improve to 0.28% and 0.901. Thus a forecast failure can support “one exponential is inadequate” while the ultimate endpoint and a mechanistic two-mode interpretation remain nonidentifiable.

10 Empirical Stress Tests

The synthetic studies establish what the diagnostics can recover under known conditions. We now ask what the same hierarchy of tests says about available self-improvement trajectories, beginning with the published evidence and then moving to a controlled reproduction and broader forecast baselines.

10.1 Published trajectories

Sun et al. validate their law with separate exponential fits to solver uncertainty, verifier uncertainty, and their gap over ten self-improvement epochs, reporting $R^2 > 0.9$ [3]. The public arXiv source contains the vector figures but no numerical trajectories, code, or seed-level data. We therefore performed a qualified artifact-level audit of the main Phi-4 training figure. Its four panels each preserve twenty vector scatter points at half-epoch spacing for all three observables, giving twelve trajectories and 240 figure-derived values.

For each trajectory, we fit (45) and (46) on the first fifteen checkpoints and forecast the last five. A 2,000-replicate parametric bootstrap calibrates D under a fitted rank-one null with independent Gaussian residuals. Rank two has lower held-out MSE on ten of twelve curves; nine reject at the uncorrected five percent level, and seven remain after Holm–Bonferroni correction across the twelve tests. The corrected rejections comprise the Math/QE solver curve and all solver, verifier, and gap curves in both TF panels.

The kernel reduction supplies a stronger check than twelve separate fits: the coupled constant-coefficient law predicts one shared mode for solver, verifier, and gap within a panel. After equal-range normalization, a common rank-one mode has higher held-out MSE than three separate rank-one modes in all four panels:

Panel	MSE factor	Holm rank-two
Math/QE	4.09	1/3
GSM8K/QE	18.55	0/3
Math/TF	10.06	3/3
GSM8K/TF	4.32	3/3

This comparison is descriptive because the plotted gap is algebraically related to solver and verifier and their seed-level covariance is unavailable. More generally, vector extraction cannot distinguish true modal structure from endpoint error, time-varying rates, aggregation across seeds, or other smooth misspecification. The result therefore improves the empirical question—it identifies a concrete failure mode and a raw-data test—without yet proving a multi-mode mechanism.

10.2 Fresh-seed reproduction

We also implemented a structural rather than numerical reproduction of the inner-clock experiment in Sun et al. [3]. The same Qwen2.5-0.5B-Instruct model acts as solver and True/False self-verifier. For each prompt it samples three candidates, filters at verifier probability 0.5, selects the qualifying candidate of smallest length-normalized NLL, and uses sixteen fixed selections as pseudo-labels for rank-16 LoRA SFT. The run uses ten epochs, learning rate 10^{-5} , and weight decay 0.01. Eight fixed held-out arithmetic prompts are evaluated before training and twice per epoch, giving twenty-one checkpoints. Three declared fresh seeds change the synthetic task sample and all sampling randomness. This scale fits a 4 GB GTX 1650; it replaces the published 3–8B models, $N = 16$, longer generations, benchmark tasks, and A800 hardware. A preliminary GSM8K calibration produced no correct candidate among thirty-two samples, motivating the controlled task.

For each seed we fit signed rank-one and rank-two residual laws on the first fifteen checkpoints and forecast the final six. We add a last-value persistence forecast: a richer exponential is not useful evidence if it merely beats a poorer exponential. The table reports pseudo-label accuracy, initial and final solver/verifier accuracy, initial and final per-token gap, the number of six observables on which rank two wins, and the number on which the better modal forecast beats persistence.

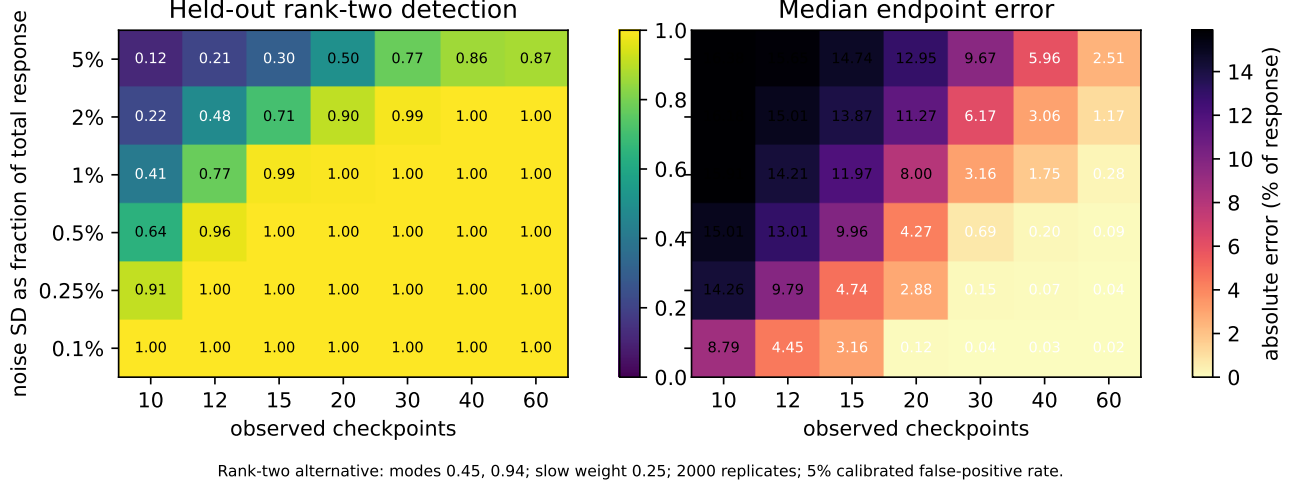


Figure 3: **Detection is easier than endpoint recovery.** Rank-one and rank-two models jointly estimate the asymptote from the training prefix and are scored on the held-out tail. The left panel gives power to prefer rank two under the alternative (42); the right gives the median absolute rank-two endpoint error as a percentage of the total response. Each condition uses 2,000 calibration, test-null, and alternative replicates. High model-selection power can occur while endpoint estimation remains poor.

Seed	PL	A_s	A_v	G	R2	M/P
20260718	.625	.50 → .875	.875 → .75	.241 → -.015	4/6	1/6
20260719	.625	.50 → .50	.625 → .625	.054 → .041	4/6	2/6
20260720	.625	.75 → .75	.875 → .875	.058 → .029	1/6	2/6

The per-token gap narrows in all three seeds, reproducing qualitative gap closure, but accuracy improvement occurs only in the first. Rank two wins nine of eighteen series–seed forecasts, exactly half; the better modal model beats persistence in only five. The per-token gap is the only observable family for which a modal model beats persistence in two seeds, and its preferred order is not stable. The stronger shared-mode check is also mixed. For total NLL, shared-rank-one/separate-rank-one MSE factors are 0.94, 0.96, 0.83; for per-token NLL they are 1.00, 0.73, 1.44. The fitted shared per-token modes 0.70, 0.60, 0.995 are not a replicated system constant.

Finally, total NLL can tell a different story from per-token NLL because the selection rule normalizes for length while the paper’s uncertainty definition does not. Correct selected responses are often longer than single solver samples, reversing the sign of the total-NLL gap. Response length and both uncertainty conventions should therefore be logged. Overall, this experiment reproduces gap closure but verifies neither a stable single exponential nor a replicated second mode. Its small evaluation panel and discrete decoding transitions make it a falsifiable pilot, not a replacement for the published scale.

10.3 Rolling forecast baselines

To test whether the modal comparison is specific and split-robust, we forecast from five prefixes containing 50, 60, 70, 75, and 80 percent of each curve. The candidate models are persistence, a training-only cross-validated local linear smoother, power-law relaxation $Y(t) = Y_\infty + c(t + \tau)^{-p}$, and signed rank-one and rank-two residuals. The resulting winner counts

are

Source	Cases	Persist.	Local	Power	R1	R2
Published	60	0	4	27	13	16
Reproduction	90	52	10	13	8	7

On the published curves, at least one modal forecast beats persistence in all sixty cases, confirming a smooth trend, but power law beats both modal models in thirty. No curve has one winner at all five boundaries; mean winner consensus is 0.53. The earlier rank-two advantage therefore rejects a rigid rank-one exponential without specifically identifying a second exponential mode. On the reproduction, the better modal model beats persistence in only 27/90 cases, persistence is the consensus winner for fifteen of eighteen trajectories, and only two trajectories keep one winner across all splits.

A fixed-seed synthetic calibration uses 200 heterogeneous rank-one and 200 rank-two curves with twenty-one checkpoints and one-percent response-scale noise. Naively choosing the lower modal forecast MSE is correct only 0.651 of the time and is biased toward rank two. Calibrating the log MSE ratio at a five-percent rank-one false-positive rate gives rank-two detection power from 0.43 to 0.625 across the five splits. Hence forecast improvement can detect misspecification without reliably classifying modal order at this resolution.

10.4 Endpoint-free inverse audit

We also apply an endpoint-free inverse diagnostic to first differences. For a candidate order R , we calibrate the squared Hankel energy beyond rank R on 400 matched synthetic nulls, then recover poles with a truncated matrix pencil. At one-percent noise the selector chooses the correct rank-two order in only 1.15% of heterogeneous rolling cases, and even when

supplied the true order the 21-checkpoint pencil recovers both poles within 0.05 in only 12.5%. Acceptance of rank one on all sixty published cases is therefore non-informative. Conditional one-pole summaries are nevertheless stable: the median gap, solver, and verifier poles are 0.740, 0.790, 0.889, respectively, with that ordering and pairwise nonoverlapping 10th–90th percentile perturbation intervals in all four panels. This supplies a direct failure of the exact shared-single-mode restriction without identifying a replacement mode count. On the reproduction, most cases have high numerical Hankel rank but no stable poles, consistent with discrete jumps rather than many physical modes.

10.5 Global, local, and dense-start dynamics

The first local-versus-global audit compares a shared rank-one response, a shared rank-two response, and a continuous two-phase exponential with broader power-law, local-linear, and persistence baselines. On the four published panels, shared rank two wins all 20 structural rolling comparisons and the two-phase model wins none against it. Once the broader baselines are included, separate power laws win 12 cases, shared rank two wins 6, separate rank one wins 1, and local linear wins 1. All 56 published local-rate windows fall in $(0, 1)$, but their panel medians range from 0.751 to 0.843 with no consistent early-to-late shift. The fitted change point drifts with the training prefix. The published curves therefore look like smooth multiscale or nonmodal relaxation, not a stable shared phase transition.

The reproduction has the opposite shape. Shared two phase wins 11/15 structural comparisons, with prefix-stable fitted change points at epochs 1.0, 1.5, and 3.5 for the three seeds; two of those fits use an extreme $0.995 \rightarrow 0.05$ rate change. Those fits describe delayed jumps into plateaus more naturally than smooth startup exponentials. Matched synthetic controls select true two-phase dynamics only 51% of the time at this checkpoint horizon. Thus the apparent local law is a useful measurement hypothesis, not an identified mechanism.

To resolve that ambiguity, seed 20260719 was rerun with evaluation after every optimizer update through epoch 4 (quarter-epoch spacing), followed by the original half-epoch schedule. The 21 shared checkpoints agree exactly with the original trajectory across all logged metrics, validating that the dense measurements did not change training. The added checkpoints localize the largest early event to one update, from epoch 1.25 to 1.5: solver total uncertainty falls from 10.416 to 2.394 while mean solver response length falls from 14.375 to 6.875 tokens; solver accuracy falls from 0.50 to 0.375, and verifier accuracy falls from 0.75 to 0.625. Solver per-token uncertainty decreases only from 0.379 to 0.333, while verifier per-token uncertainty increases from 0.327 to 0.410. A later verifier transition from epoch 3.0 to 3.25 similarly coincides with a length increase from 14.125 to 18.25 tokens and a verifier-accuracy decrease. These are behavioral and measurement transitions, not clean evidence of a capability takeoff. They also show why a general kernel remains useful when a finite modal law fails: a concentrated or signed response can be measured without pretending that it is a sum of stable exponentials.

11 Experimental Implications

The immediate experiment is a joint null test, not an unconstrained search for complexity. For each model–task–verifier trajectory family:

1. estimate or profile over plausible asymptotes;
2. fit one shared λ to normalized U_s, U_v, G trajectories;
3. test kernel-ratio and Hankel-minor residuals;
4. compare against regularized finite-modal, diffuse-spectrum, phase-changing, and nonparametric alternatives;
5. evaluate by held-out checkpoints and stability across seeds.

The synthetic studies separate two design targets. Roughly fifteen checkpoints can detect held-out rank-one failure for this alternative at one percent response-scale noise, but endpoint and mode estimation require closer to forty–sixty. Kernel-Hankel detection is more demanding because differencing attenuates slow modes and amplifies observation noise. These are conditional floors, not general prescriptions: the required count rises as the slow-mode weight or modal separation shrinks. In practice both residual and kernel analyses should be reported, with asymptote sensitivity for the former and noise-aware window selection for the latter. Raw data should retain paired seed-level solver, verifier, and gap measurements so that the shared-mode null can be tested with the correct covariance. If the available checkpoint count cannot distinguish one mode from two in a matched synthetic study, the correct conclusion is nonidentifiability at the available resolution, not evidence for rank one.

The resulting narrative is deliberately narrower than a general theory of self-improvement. The standard solver–verifier law supplies a useful null; the takeoff-kernel reduction states exactly what that null predicts; and finite modal, diffuse-spectrum, and time-local models supply controlled alternatives when it fails. Current evidence is strong enough to question one exact global shared exponential rate, but not strong enough to name a discrete mode count, to assert a universal local exponential, or to separate causal sources from their aggregate response. Better checkpoint density, independent observables, paired seed-level trajectories, and explicit asymptote and response-length sensitivity are the next research requirements.

References

- [1] A. Huang, A. Block, D. J. Foster, et al. Self-improvement in language models: The sharpening mechanism. *ICLR*, 2025. arXiv:2412.01951.
- [2] Y. Song, H. Zhang, C. Eisenach, S. Kakade, D. Foster, and U. Ghai. Mind the gap: Examining the self-improvement capabilities of large language models. *ICLR*, 2025. arXiv:2412.02674.
- [3] Y. Sun, Y. Liang, Z. Zhang, X. Liu, and J. Teng. Theoretical modeling of large language model self-improvement training dynamics through solver–verifier gap. *ICLR*, 2026. arXiv:2507.00075.

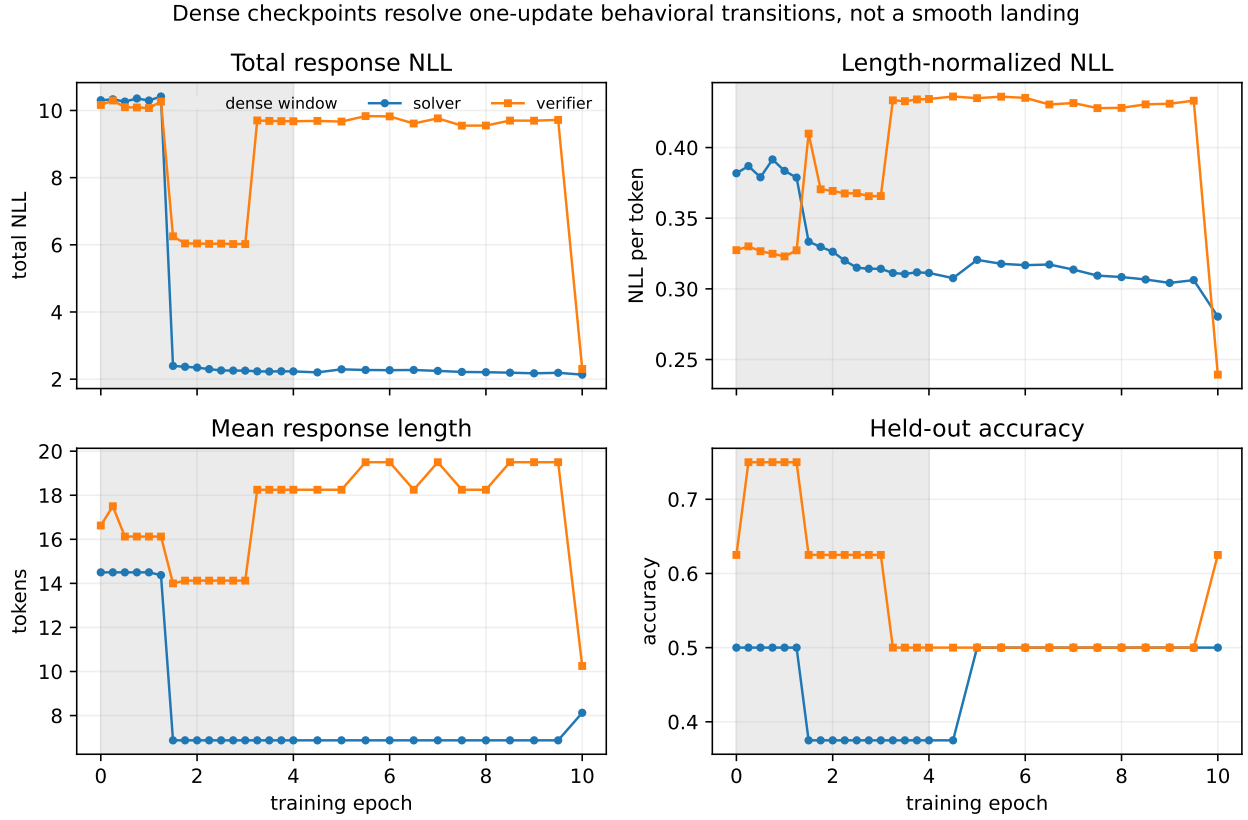


Figure 4: **Dense-start audit for seed 20260719.** Quarter-epoch measurements through epoch 4 (shaded) resolve one-update transitions in total and per-token uncertainty, response length, and accuracy. The sharp changes are coupled to decoding behavior and do not form a smooth exponential landing.